

PRD

Drinking Diary

Product Requirements Document (PRD)

Version: 1.0

Platform: iOS

Product Type: Private Lifestyle Tracking App

1. Product Overview

1.1 Product Vision

Drinking Diary is a private alcohol tracking application that allows users to record their drinking activities, learn about different alcoholic beverages, and predict their intoxication level based on personal drinking history and physiological data.

The app aims to combine:

- alcohol knowledge database
- personal drinking diary
- intoxication prediction
- AI-generated drinking insights

into a single lifestyle tool for alcohol enthusiasts.

1.2 Product Positioning

Drinking Diary is primarily a:

Personal Alcohol Diary + Alcohol Knowledge App

Core value propositions:

1. Track personal drinking history
2. Understand alcohol types and cocktail recipes
3. Predict intoxication levels using BAC and personalized models
4. Generate AI-powered drinking reports and recommendations

The app is **private**, meaning data is only visible to the user.

2. Target Users

Primary Users

Alcohol enthusiasts who:

- enjoy cocktails, beer, whiskey, wine, or spirits
- want to track their drinking habits
- are curious about alcohol knowledge
- want to better understand their alcohol tolerance

Typical scenarios:

- recording a night out drinking
 - learning about cocktails
 - estimating intoxication level before drinking
 - reviewing drinking history
-

3. Product Goals

Goal 1

Provide an easy way for users to log their drinking activity.

Goal 2

Build a personal alcohol consumption profile.

Goal 3

Predict intoxication level based on drinking behavior and physiological information.

Goal 4

Help users explore alcohol knowledge and bar locations.

4. Core Features

The product consists of **four main modules**.

4.1 Drink Diary

Users can record their drinking activities through a calendar-based logging system.

The diary allows users to track drinking events and associated contextual information.

Recorded Data per Drinking Session

```
date
drinks
location
companions
mood
notes
intoxication level (slider)
```

Example:

```
Date: March 6
```

```
Drinks:
IPA x2
```

Grey Goose Vodka x1

Location: Atlas Bar

Mood: Happy

With: Alex

Drunk Level: 6 / 10

Each logged drink is linked to a drink entry that contains alcohol information (ABV, category, etc.), which is used for BAC and intoxication prediction.

4.1.1 Drink Search Autocomplete

To simplify drink logging and improve data accuracy, the app provides an intelligent search suggestion system when users enter drink names.

As the user types in the drink input field, the system automatically displays a dropdown list of possible drink matches.

This allows users to quickly select the correct drink brand or type.

Trigger Behavior

The autocomplete suggestion system activates when the user enters **at least two characters** in the drink search field.

Example input:

vod

Suggested results:

Grey Goose · Vodka · 40%

Ketel One · Vodka · 40%

Absolut · Vodka · 40%

Smirnoff · Vodka · 40%

Selecting a suggestion automatically fills in drink metadata, including alcohol category and ABV.

Matching Logic

Drink suggestions are ranked using the following criteria:

1. Prefix matching (highest priority)
 2. Partial text matching
 3. Fuzzy matching (handles typos)
 4. User preference weighting (frequently logged drinks ranked higher)
-

Data Sources

The autocomplete system retrieves drink suggestions from three sources in priority order:

1. **User custom drinks** (moonshine or personal drinks)
2. **Recently logged or favorite drinks**
3. **Official alcohol database**

This ensures that drinks created by the user appear quickly in search suggestions.

Manual Entry Option

If the drink the user is searching for does not exist in the database, the user can manually add it as a custom drink.

Example UI:

```
No results found
+ Add Custom Drink
```

After creation, the custom drink becomes part of the user's personal drink library and will appear in future autocomplete suggestions.

4.2 Alcohol Knowledge Database (Updated)

The app provides an alcohol knowledge database that allows users to explore different types of alcoholic beverages.

The database includes both **cocktails and base alcohol categories**, enabling users to learn about drink ingredients, alcohol content, flavor profiles, and origin stories.

Supported categories include:

```
cocktails
beer
whiskey
gin
rum
vodka
tequila
wine
liqueur
```

Users can browse drinks by category or search directly by name.

Data Sources

To provide comprehensive alcohol information, the system integrates multiple data sources.

1. Cocktail Database

Cocktail recipes and preparation methods are retrieved from:

```
TheCocktailDB API
```

This database provides:

- cocktail names
- ingredients

- preparation instructions
 - glass types
 - alcohol categories
-

2. Alcohol Brand and Category Database

Information about base alcohol types and brands (e.g., beer, vodka, whiskey) is retrieved from an alcohol brand database.

This database provides:

```
drink name  
brand  
alcohol category  
ABV  
flavor profile  
country of origin
```

Examples:

```
Grey Goose – Vodka – 40% – France  
Guinness – Stout Beer – 4.2% – Ireland  
Macallan 12 – Whisky – 40% – Scotland
```

This database is used for:

- drink search autocomplete
 - BAC calculation
 - alcohol knowledge browsing
-

3. User Custom Drinks

Users may create custom drink entries when a drink is not found in the database.

These include:

```
moonshine  
homemade alcohol  
rare local alcohol  
custom cocktails
```

Custom drinks are stored in the user's **Personal Drink Library** and will appear in drink search suggestions.

Database Structure Update

To support multiple alcohol categories, the **Drinks table** includes the following fields:

```
id  
name  
brand  
category  
abv  
volume  
origin_country  
flavor_profile  
aliases
```

Example record:

```
name: Grey Goose  
brand: Grey Goose  
category: Vodka  
abv: 40  
origin_country: France
```

Integration With Drink Logging

All drinks in the database (cocktails, alcohol brands, and custom drinks) are searchable through the drink autocomplete system.

When a drink is selected:

```
drink name
category
ABV
volume
```

are automatically populated and used in **BAC and intoxication prediction calculations**.

4.3 Intoxication Prediction

One of the core differentiating features.

The app predicts intoxication levels based on:

- physiological data
 - drink type
 - drinking history
-

Model Structure

The prediction system is a **hybrid model**.

Layer 1: Physiological BAC Model

The app calculates Blood Alcohol Concentration using the Widmark Formula.

BAC formula:

$$\text{BAC} = (A \times 5.14 / (W \times r)) - 0.015 \times H$$

Where:

A = alcohol consumed (ounces)

W = body weight

r = body water constant

```
male = 0.73  
female = 0.66
```

H = hours since drinking started

Layer 2: Personalized Intoxication Model

Because subjective intoxication differs between individuals, the system also estimates:

```
drunk level (0–10)
```

Inputs include:

```
BAC  
drink count  
drink type  
drinking speed  
user drinking history  
body data
```

Output:

```
predicted intoxication level
```

The model gradually improves as more user data is collected.

4.4 Bar Map

Users can explore and save bars.

Integration:

Google Places API

Users can:

- view bars on map
- mark visited bars
- save favorite bars

The app generates:

```
My Drinking Map
```

which visualizes the user's drinking locations.

4.5 AI Features

Two AI-powered features will be included.

AI Drinking Report

The system generates weekly summaries of drinking behavior.

Example:

```
Weekly Drinking Summary
```

```
You had 6 drinks this week.
```

```
Most frequent alcohol:
```

```
IPA
```

```
Favorite bar:
```

```
Atlas Bar
```

```
Highest intoxication night:
```

```
Friday (7/10)
```

Reports may also include insights such as:

- drinking patterns
 - favorite drinks
 - monthly drinking trends
-

AI Bar Recommendation

Based on:

favorite alcohol types
visited bars
location
drinking preferences

The system recommends a bar each week.

Example:

Recommended bar this week:
Employees Only (Singapore)

5. App Structure

The application uses a **5-tab interface**.

Home
Diary
Explore
Map
Profile

Home

Displays:

- today's drinks
 - predicted intoxication level
 - BAC estimate
 - weekly drink count
 - recommended bar
-

Diary

Calendar view of drinking history.

Users can:

- view past drinking sessions
 - add new drink logs
 - record intoxication levels
-

Explore

Alcohol database.

Users can:

- search drinks
 - view recipes
 - learn about alcohol types
-

Map

Bar exploration map.

Functions:

- discover bars
 - save bars
 - mark visited bars
-

Profile

User personal data:

```
height
weight
gender
drinking tolerance
```

Also includes:

```
favorite drinks
lifetime drink count
monthly reports
```

6. Data Structure

Key database tables.

Users

```
id
height
weight
gender
tolerance
```

Drinks

```
id
name
type
```

```
abv  
volume
```

DrinkLogs

```
user_id  
drink_id  
amount  
time  
date
```

Sessions

A drinking event.

```
start_time  
end_time  
location  
mood  
companions  
notes  
drunk_level
```

Bars

```
name  
location  
rating
```

Favorites

```
user_id  
drink_id
```

7. Technology Stack

The Drinking Diary application is designed as an **iOS-only mobile application** with a lightweight backend architecture.

The system integrates external APIs for drink information, location services, and AI-powered insights.

The technology stack prioritizes **stability, fast search performance, and simple deployment** for a small private user base.

7.1 Frontend

The mobile application will be developed using:

```
SwiftUI
```

SwiftUI provides:

- modern declarative UI framework
- fast development iteration
- strong integration with Apple ecosystem
- efficient state management for interactive components such as drink search autocomplete

Key UI components include:

- drink logging interface
- drink search autocomplete dropdown
- calendar-based drinking diary
- alcohol knowledge browsing

- map-based bar exploration
-

7.2 Backend

The backend system will be implemented using:

Supabase

Supabase provides:

- authentication
- PostgreSQL database
- REST and realtime APIs
- cloud storage
- simple deployment

The backend stores:

- user profiles
- drink logs
- drinking sessions
- custom drinks
- user preferences
- drinking statistics

Supabase enables secure user data storage while keeping the architecture lightweight.

7.3 Local Data Storage

To ensure fast search performance and offline functionality, some data will be cached locally on the device.

Local storage may include:

```
drink brand database
recently logged drinks
favorite drinks
custom drinks
```

Local storage can be implemented using:

```
SQLite
CoreData
```

Local caching improves performance for the **drink search autocomplete system**.

7.4 External APIs

The application integrates several external APIs to provide drink information, location services, and AI-powered features.

7.4.1 Cocktail Database API

Cocktail recipes and preparation information are retrieved from:

```
TheCocktailDB API
```

This API provides:

- cocktail names
- ingredients
- preparation instructions
- glass types
- drink categories

This data powers the **cocktail knowledge section** in the Explore module.

7.4.2 Alcohol Brand Database

To support a wide range of alcoholic beverages, the system maintains a database of alcohol brands and drink categories.

Supported categories include:

```
beer  
whiskey  
vodka  
gin  
rum  
tequila  
wine  
liqueur
```

This database contains information such as:

```
drink name  
brand  
alcohol category  
ABV  
origin country  
flavor profile  
aliases
```

Example entries:

```
Grey Goose – Vodka – 40% – France  
Guinness – Stout Beer – 4.2% – Ireland  
Macallan 12 – Whisky – 40% – Scotland
```

This database is used for:

- drink search autocomplete
- drink logging

- BAC calculation
 - alcohol knowledge browsing
-

7.4.3 Map and Bar Location API

Bar discovery and map visualization are powered by:

Google Places API

This API provides:

- bar locations
- ratings
- opening hours
- geographic coordinates

These data are used in the **Map module**, where users can:

- discover bars
 - mark visited bars
 - save favorite bars
-

7.4.4 AI Services

AI functionality is powered by:

OpenAI API

AI services are used for:

- generating weekly drinking reports
- generating drinking insights
- recommending bars based on user preferences

Example output:

Weekly Drinking Summary

You had 6 drinks this week.

Most frequent alcohol: IPA

Favorite bar: Atlas Bar

Highest intoxication: Friday night (7/10)

7.5 Search and Autocomplete System

The drink search input field includes an **autocomplete suggestion system** to simplify drink logging.

When users type part of a drink name, the system displays a list of suggested drinks.

Suggestions are retrieved from:

1. user custom drinks
2. recently logged drinks
3. official alcohol database

The system ranks results based on:

- prefix matching
- partial text matching
- fuzzy matching
- user preference weighting

If a drink is not found, users can create a **custom drink entry**, which will then be included in future search suggestions.

7.6 AI Recommendation Engine

The system includes a lightweight AI recommendation module.

Inputs include:

```
favorite alcohol types
drinking history
visited bars
user location
```

Outputs include:

```
weekly drinking report
personal drinking insights
bar recommendations
```

This module enhances user engagement by providing personalized drinking insights.

7.7 Future Scalability

The architecture is designed to support additional integrations in the future.

Potential integrations include:

```
Untappd API (beer database)
Vivino API (wine database)
spirits and whisky databases
```

These integrations would expand the alcohol knowledge database.

7.8 Security and Privacy

Because Drinking Diary is a **private personal tracking application**, user data is protected using:

```
secure authentication
encrypted data transmission
```

private user data storage

User drinking records are not shared publicly.

9. MVP Scope

The first version of Drinking Diary will focus on the **core drink tracking and alcohol knowledge experience**, while more advanced features will be introduced in later versions.

Phase 1 — Core Drinking Diary

Core functionality for recording drinking activity.

Features include:

- drink logging
 - drink search autocomplete (for logging drinks)
 - custom drink creation
 - intoxication slider
 - calendar-based drink history
-

Phase 2 — BAC Prediction

Basic intoxication prediction system.

Features include:

- BAC calculation using Widmark formula
- estimated intoxication level display

Required inputs:

weight
gender

drink ABV
drink volume
drinking duration

Phase 3 — Alcohol Knowledge Database

Alcohol information browsing.

Features include:

- cocktail knowledge database integration
- alcohol category browsing
- drink information pages

Phase 4 — Bar Map

Location-based bar discovery.

Features include:

- map view
- bar discovery
- favorite bars
- visited bars

Powered by:

Google Places API

Phase 5 — AI Insights

AI-generated drinking insights.

Features include:

- weekly drinking report
- basic bar recommendation

Powered by:

OpenAI API

MVP Release Goal

The MVP release should enable users to:

- record drinking activities
- estimate intoxication level
- explore alcohol knowledge
- review drinking history

Advanced features such as **improved AI recommendations and expanded alcohol databases** will be introduced in later versions.

9. Privacy

The app is designed as a **private personal diary**.

User data:

- is not public
- is not shared with other users

No social features will be included in the initial version.

10. Success Metrics

Key metrics include:

- number of drink logs per user

- weekly active users
- user retention
- frequency of report generation
- number of saved drinks